l=0. i.e. it occupies 1s orbital. The plot of $R(r)^2$ versus r for 1s orbital is given in Figure 2.3

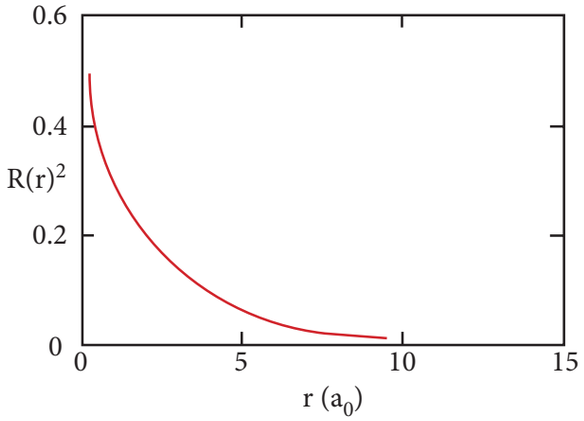

Figure. 2.3 Plot of $R(r)^2$ versus r for 1s orbital of hydrogen

The graph shows that as the distance between the electron and the nucleus decreases, the probability of finding the electron increases. At r=0, the quantity $R(r)^2$ is maximum i.e. The maximum value for $|\Psi|^2$ is at the nucleus. However, probability of finding the electron in a given spherical shell around the nucleus is important. Let us consider the volume (dV) bounded by two spheres of radii r and r+dr.

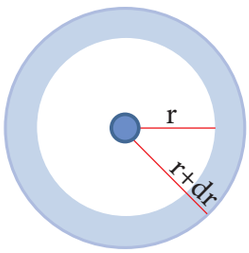

Figure 2.4

Volume of the sphere, $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dr} = \frac{4}{3}\pi(3r^2)$$

$$dV = 4\pi r^2 dr$$

$$\Psi^2 dV = 4\pi r^2 \Psi^2 dr \quad \text{---------- (2.16)}$$

The plot of $4\pi r^2 . R(r)^2$ versus r is given below.

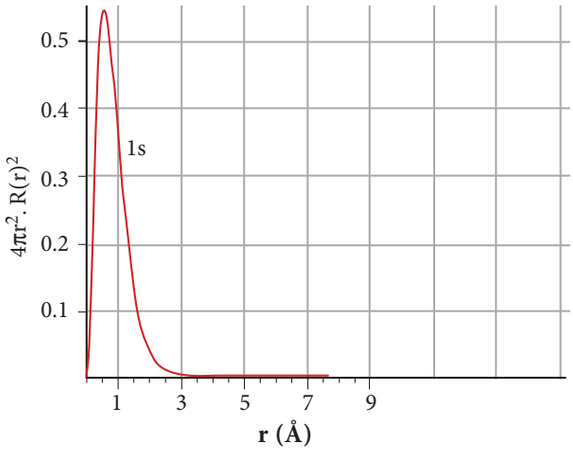

Figure 2.5 Plot of $4\pi r^2 . R(r)^2$ versus r for 1s orbital of hydrogen

The above plot shows that the maximum probability occurs at distance of 0.52 **Å** from the nucleus. This is equal to the Bohr radius. It indicates that the maximum probability of finding the electron around the nucleus is at this distance. However, there is a probability to find the electron at other distances also. The radial distribution function of 2s, 3s, 3p and 3d orbitals of the hydrogen atom are represented as follows.

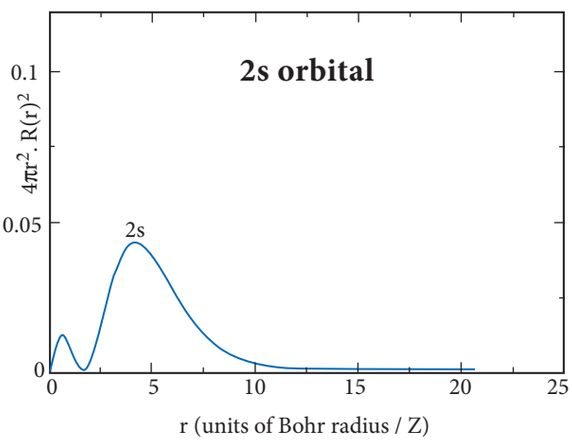

Figure 2.6 (a) - Plot of $4\pi r^2 . R(r)^2$ versus r for 2s orbitals of hydrogen

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